



BAHIR DAR UNIVERSITY
BAHIR DAR INSTITUTE OF TECHNOLOGY
FACULTY OF MECHANICAL AND INDUSTRIAL
ENGINEERING
MECHANICAL ENGINEERING PROGRAM

Course Guidebook for Fluid Mechanics

General Information of The course

Course Name:	Fluid Mechanics
Course Code:	MEng 3171
Course Credits Points:	5 hrs
Contact Hours:	3 hrs lecture, 2 hrs Tutorial and 2 hrs laboratory/Practice per week
Course Module:	Thermo-Fluid Engineering
Course Category:	Major
Status of Course:	Compulsory
Pre-Requisites:	Engineering Mechanics II and Applied Mathematics III
Students' Department:	Mechanical Engineering
Degree Program:	Mechanical Engineering
Students' Admission:	Regular
Year of Study:	3
Semester:	First Semester
Academic Year:	2025/2026
Course Duration:	From sept 2025 to Jan 2025
Course Chair:	Dr. Bimerewu Tamirat
Course Instructors:	Tereche Getnet, MSc in Thermal Engineering Email: terecheg83@gmail.com Office: B-8: No.6
Consultation:	Monday morning 11:30 AM -12:30 AM wednesday afternoon 4:00 PM-11:30 PM

OBJECTIVES OF THE COURSE

Students at the end of this course will be able to:

1. Permits students to understand the basic concepts, principles, laws, observations, and models of fluids at rest and in motion,
2. Provides basis for understanding fluid behavior for engineering design and control of fluid systems,
3. Develop competence with mass, energy and momentum balances for determining resultant interactions of flows and engineered and natural systems,

4. Develop bases for correlating experimental data, designing procedures, and using scale models of fluid flows,
5. Know Newtonian and no- Newtonian flows, enables students to learn the nature of rotation, circulation, resistance (viscous, turbulent), boundary layers, and separation with applications to drag and lift on objects,
6. Learn methods for computing head losses and flows in simple pipe and channels.
7. Identify, formulate and solve engineering problems involving compressible fluid flows
8. Understand the principles of operation of flow measuring instruments, conduct measurements, evaluate the data and draw conclusions

Course Learning Outcomes (CLO)

At the end of the course, students will be able to:

No	Course Learning Outcome (CLO)	Performance Indicators (PIs)	Program Student Outcome (PSO)
1	Determine fluid properties, measure fluid characteristics	A1, A2, B1	1,2
2	Understand flow types, flow conditions and flow theory	A1, A2, B1	1,3
3	Understand turbulence flow and laminar flow effects on real flow	B2	2
4	Know analysis and design of flat slabs for flexure, shear and check for deflection requirement	B2, B4	2
5	Describe the behavior of members with torsion and design elements for torsion	C1	3
6	Analyze and design of reinforced concrete staircases	D5, D6	4

The Course Time Schedule and Teaching Methodology

The following topics will be covered in this course within this semester.

Table 1. Time Schedule and Teaching Methodology of the course

Time Schedule	Course Contents	Teaching Methodology and References	Attaining student outcome
Week 1&2	UNIT 1 Introduction 1. Relevance and Significance In Engineering Applications 2. Definitions 3. Fluid Properties 4. Flow Analysis Techniques 5. Flow Patterns	Reference 5	Outcome 6
Week 3,4&5	UNIT 2 Fluid Statics 1. Introduction 2. Pressure Specifications 3. Hydrostatic Pressure Distributions 4. Manometer 5. Hydrostatic Forces on Plane Surfaces 6. Hydrostatic Forces on Curved Surfaces 7. Buoyancy and Stability 8. Pressure Variation with Rigid Body Motion	Reference 4&3	Outcome 3
Week 6&7	UNIT 3 Integral Relations for A Control Volume 1. Introduction 2. Physical Laws of Fluid Mechanics 3. The Reynolds Transport Theorem 4. Conservation of Mass Equation 5. Linear Momentum Equation 6. Angular Momentum Equation 7. Energy Equation 8. Bernoulli Equation	Reference 5&3	Outcome 1
Week 8&9	UNIT 4 Differential Relations for A Fluid Flow 1. Introduction 2. Acceleration Field 3. Conservation of Mass Equation 4. Linear Momentum Equation 5. Energy Equation 6. Boundary Condition 7. Stream Function 8. Vorticity and Irrotationality	Reference 5	Outcome 3

Week 10&11	UNIT 5 Dimensional Analysis and Similitude 1. Introduction 2. Dimensional Homogeneity 3. Buckingham Pi Theorem 4. Non Dimensionalization of Basic Equations 5. Similitude 6. Significance of Non-Dimensional Numbers In Fluid Flows	References	Outcome 6
Week 13&14	UNIT 6 Boundary Layer Concept 1. Introduction 2. Reynolds Number and Geometry Concept 3. Momentum Integral Equations 4. Boundary Layer Equations 5. Flow Over A Flat Plate 6. Flow Over Cylinder 7. Pipe Flow Fully Developed Laminar Pipe Flow Turbulent Pipe Flow Losses in Pipe Flow	Reference 3	Outcome 4&5
Week 15	UNIT 7 Compressible Flow 1. Introduction 2. Speed of Sound 3. Steady Flow 4. Flow with Area Change Nozzles and Diffusers Normal Shock Wave Duct Flow with Friction	Reference 3,4&5	Outcome 1&2
Week 16	UNIT 8 Introduction to 2D-Potential Flow Theory 1. Introduction 2. Plane Potential Flow 3. Superposition of Plane - Flow Solutions - Plane Flow Past Closed-Body Shapes 4. Aero foil Theory		Outcome 3

Assessment and Evaluation

The course utilizes a variety of continuous assessment and evaluation methods to ensure the successful achievement of course learning outcomes. These methods include assignments, projects, quizzes, midterm exams, final exams, site visit reports, homework, classwork, presentations, and more. It is imperative that all assignments and tasks are completed within the specified deadlines. If you encounter any difficulties with the assignments or tests, please contact the instructor promptly for assistance.

Home works, Assignments and project: Assignments may involve collaborative work for brainstorming and developing general approaches. However, unless specified otherwise, each individual must complete their own submissions. Assignments and projects may entail both individual and group components, with equal participation expected from all group members. While collaboration is encouraged, submitting another individual's work as one's own constitutes academic dishonesty and will incur in penalties. All group members must contribute equally. Timely submission is mandatory; as late submissions will not be graded. Oral exams will accompany assignments and projects, requiring thorough preparation from all members. Additionally, the instructor reserves the right to assign additional tasks or assignments as necessary.

Quizzes: Quizzes serve to assess students' current understanding of the coursework. They are conducted in class following lecture or tutorial sessions.

Mid-term Exam and Final Exam: The exams will be closed book. During exam all necessary materials will be provided by examiner.

Table 2. Summary of Assessments

No	Assessment type	Mark allotted	Content Coverage	Attaining course learning outcome (CLO)	Time Schedule
1	Assignment 1	10	Chapter 1&2	1	Week 3
4	Lab	5	Chapter 1,2 &3	2,5,6	Week 4
7	Mid exam	25%	Chapter 1-3	1,3,4,6	Week 8
8	Assignment 2(quiz)	10	Chapter 4	4,2&1	Week 10
13	Final Exam	50%	Chapter 5,6,7&8	1,2,3,4,5,6	Week 16
	Total	100%			

Laboratory/Workshop/ session

Table 3. Summary of laboratory /Work shop/ session content and required material

No	Schedule	Laboratory Topic	Short Description of laboratory	Attaining course outcome
1	Week 1	1. Measurement of dispersion around turbulent jet 2. Measurement of velocity profile and boundary layer growth over a flat plate-effect of smooth/rough surface and favorable/adverse pressure gradient		2
2	Week 2	3. Verification of Bernoulli's equation		3
3	Week 3	4. Comparison of losses in nozzle and diffuser type duct flows		4

4	Week 4	Measurements on Free vortex flow and Forced Vortex flow		5
	Week 5	Determination of CD and comparison of CD for Orifice and Venturimeter		6
	Week 6	Reynolds's experiment		4
	Week 7	Measurement of drag and lift of an aero foil at different angles of attack		2
	Week 8	. Evaluation systems of the variance of lift and Drag on an aero foil via flaps and slats		4,3

COURSE POLICY

All students are expected to adhere to the code of conduct for students as per the Senate Legislation of Bahir Dar University, December 2020, during the duration of this course. Academic misconduct, which includes cheating, fabrication, and plagiarism, will not be tolerated and will be promptly reported to the relevant authorities for appropriate action.

The teaching methodology for this course will be diverse, encompassing activities such as reading, lectures, tutorials, construction site visit, discussions, homework, classwork exercises, and presentations, etc. Active participation from students is expected, including asking questions and raising issues whenever clarification is needed.

Regular attendance and punctuality are expected from all students. Those who miss more than 15% of the semester's classes will not be eligible to sit for the final exam. Attendance will be recorded for each class, and students are responsible for all announcements and changes made in class. Any student who exceeds the 15% absence threshold will receive a grade of 'NG' (No Grade) and must provide acceptable reasons for their absence, as per Senate legislation, Art. 174 for NG management.

Cell phones brought to class must be set to silent mode to avoid disruption to the teaching-learning process. Any form of disturbance during class time is unacceptable and must be avoided.

All students are required to have complete records of course assessments and evaluations. If a student misses a quiz, midterm exam, or any other continuous assessment without a valid reason, they will receive a score of zero. However, if a student fails to attend the final exam or complete a term paper/project without a valid reason, the instructor will record an "NG" according to senate legislation, Art. 171.

It is important to carefully follow the instructions provided in your course guidebook for completing all requirement of the course.

Grading system

The grading scale of the course is fixed scale according to senate legislation of Art. 177.

Table 4. Grading scale of the course

Raw Mark Interval (100%)	Corresponding Letter Grade	Corresponding Fixed Number Grade
$90 \leq x \leq 100$	A ⁺	4.0
$85 \leq x < 90$	A	4.0
$80 \leq x < 85$	A ⁻	3.75
$75 \leq x < 80$	B ⁺	3.50
$70 \leq x < 75$	B	3.00

$65 \leq x < 70$	B ⁻	2.75
$60 \leq x < 65$	C ⁺	2.50
$50 \leq x < 60$	C	2.00
$45 \leq x < 50$	C ⁻	1.75
$40 \leq x < 45$	D	1.00
$x < 40$	F	0.00

Text book:

Frank M. White, Fluid Mechanics with Student CD (McGraw-Hill Series in Mechanical Engineering), Oct 17, 2006.

References:

1. Anthony Esposito, “Fluid Power with Applications”, Pearson Education 2000
2. Fox R.W., and McDonald A.T., Introduction to Fluid Mechanics
3. Munson B.R, Fundamental of Fluid Mechanics
4. V.L.Streeter, Fluid Mechanics
5. White F.M., Fluid Mechanics

Aproval

The following bodies confirmed the course guidebook accordingly.

Position	Name	Signature	Date
Course Instructors	Tereche Getnet		
Course Module Chair			
Chair Holder (Structural Engineering)	Dr. Bimerewe Tamerat		
V/Dean			

Notes:

Student outcomes and course learning outcomes are related but distinct concepts in education:

Student Outcomes: These are the specific knowledge, skills, or attributes that students are expected to demonstrate or achieve by the end of a program or course. They focus on what students will be able to do or demonstrate as a result of their learning experiences. Student outcomes are often aligned with broader educational goals and may include abilities such as critical thinking, problem-solving, effective communication, and subject-specific skills.

Course Learning Outcomes: These are the specific statements that describe what students should be able to know or do as a result of completing a particular course. Course learning outcomes are more specific and focused on the content and objectives of the course itself. They outline the expected achievements or competencies that students should attain by the end of the course. These outcomes guide course design, instruction, assessment, and evaluation.

In summary, student outcomes are broader and pertain to the overall goals of an educational program, while course learning outcomes are more specific and relate directly to the content and objectives of individual courses within that program.

=====

In education curriculum design, both student outcome performance indicators (PIs) and course learning outcomes (CLOs) are important components, but they serve slightly different purposes:

Student Outcome Performance Indicators (PIs):

PIs are specific, measurable criteria used to assess students' attainment of learning outcomes.

They provide detailed descriptions of what students should be able to demonstrate or accomplish to meet the learning outcomes.

PIs often include observable behaviors, skills, or knowledge that can be assessed through various methods such as exams, assignments, projects, presentations, or observations.

PIs help educators and curriculum designers develop assessment strategies and tools to evaluate students' progress and achievement.

PIs are often aligned with course objectives and standards to ensure that assessment activities effectively measure student learning.

Course Learning Outcomes (CLOs):

CLOs are broad statements that describe what students are expected to learn or achieve by the end of a course or program.

They represent the overarching goals and purposes of the course and provide a framework for instructional planning and assessment.

CLOs articulate the knowledge, skills, abilities, or attitudes that students should develop as a result of their learning experiences.

CLOs are typically derived from broader educational goals, program objectives, or professional standards.

CLOs guide the selection of instructional content, activities, and assessments to ensure that they are aligned with desired learning outcomes.

In summary, PIs are the specific criteria used to measure student achievement of CLOs. While CLOs provide a high-level overview of desired learning outcomes for a course or program, PIs offer more detailed, observable indicators of student performance that can be used to assess progress and attainment of those outcomes.